

```
install.packages("sqldf")
```

Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

```
library(sqldf)
```

Loading required package: gsubfn

Loading required package: proto

Warning message:

“no DISPLAY variable so Tk is not available”

Loading required package: RSQLite

```
library(readr)
```

```
customers<-read_csv("customers.csv")
```

Rows: 650 Columns: 9

— Column specification —

Delimiter: ","

chr (5): customer_id, home_zone, customer_type, preferred_channel, account_

dbl (3): age, loyalty_score, app_engagement_score

dtm (1): signup_date

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
head(customers)
```

A tibble: 6 × 9

customer_id	age	home_zone	customer_type	signup_date	loyalty_score	app
<chr>	<dbl>	<chr>	<chr>	<dtm>	<dbl>	
C0001	26	North	SME	2024-11-27 04:25:00	44.9	
C0002	61	AIRPORT	Consumer	2025-10-28 01:04:00	55.4	
C0003	66	East	Consumer	2025-07-02 03:23:00	75.9	
C0004	75	CENTRAL	Consumer	2025-08-19 01:58:00	32.5	
C0005	26	Riverside	Consumer	2025-06-03 06:02:00	55.9	
C0006	41	WEST	Consumer	2024-03-29 13:26:00	39.9	

```
colnames(customers)
```

```
'customer_id' · 'age' · 'home_zone' · 'customer_type' · 'signup_date' · 'loyalty_score' ·  
'app_engagement_score' · 'preferred_channel' · 'account_status'
```

```
complaints<-read_csv("complaints.csv")
```

Rows: 320 Columns: 10

— Column specification —

Delimiter: ","

chr (7): complaint_id, customer_id, order_id, complaint_type, channel, severe,

dbl (2): resolution_days, compensation_amount

dtm (1): created_at

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
head(complaints)
```

A tibble: 6 × 10

complaint_id	customer_id	order_id	complaint_type	channel	severity	creat
<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<
CP0001	C0464	O00814	AppIssue	App	High	2025-02
CP0002	C0056	O00628	MissedPickup	Phone	Medium	2024-10
CP0003	C0469	O00384	Delay	Chatbot	High	2024-15
CP0004	C0631	O00406	Delay	App	Medium	2025-13
CP0005	C0535	O00154	Delay	Email	Medium	2024-05
CP0006	C0096	O00147	Delay	App	Medium	2024-07

```
colnames(complaints)
```

```
'complaint_id' · 'customer_id' · 'order_id' · 'complaint_type' · 'channel' · 'severity' · 'created_at' ·  
'status' · 'resolution_days' · 'compensation_amount'
```

```
deliveries<-read_csv("deliveries.csv")
```

Rows: 950 Columns: 13

— Column specification —

Delimiter: ","

chr (6): delivery_id, order_id, driver_id, vehicle_id, hub_id, delivery_stat

dbl (5): route_distance_km, manual_route_override_count, proof_of_completio.

dtm (2): dispatch_time, delivery_completed_at

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this mess:

```
head(deliveries)
```

delivery_id	order_id	driver_id	vehicle_id	hub_id	dispatch_time	delivery
<chr>	<chr>	<chr>	<chr>	<chr>	<dtm>	
DL00001	O00938	D004	V056	H05	2024-06-18 10:57:00	2024
DL00002	O00004	D138	V007	H02	2025-01-11 18:45:00	2025
DL00003	O00639	D006	V049	H02	2025-06-02 20:39:00	2025
DL00004	O00313	D116	V055	H02	2024-03-08 23:31:00	2024
DL00005	O00844	D108	V034	H01	2025-09-21 11:43:00	2025
DL00006	O00029	D037	V098	H03	2024-09-11 12:40:00	2024

```
colnames(deliveries)
```

```
'delivery_id' · 'order_id' · 'driver_id' · 'vehicle_id' · 'hub_id' · 'dispatch_time' ·  
'delivery_completed_at' · 'delivery_status' · 'route_distance_km' ·  
'manual_route_override_count' · 'proof_of_completion_missing' ·  
'customer_rating_post_delivery' · 'fuel_or_charge_cost'
```

```
complaints<-read_csv("complaints.csv")
```

Rows: 320 Columns: 10

— Column specification —

Delimiter: ",",

chr (7): complaint_id, customer_id, order_id, complaint_type, channel, severe,

dbl (2): resolution_days, compensation_amount

dtm (1): created_at

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
head(complaints)
```

A tibble: 6 × 10

complaint_id	customer_id	order_id	complaint_type	channel	severity	creat
<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<
CP0001	C0464	O00814	AppIssue	App	High	2025-02
CP0002	C0056	O00628	MissedPickup	Phone	Medium	2024-10
CP0003	C0469	O00384	Delay	Chatbot	High	2024-15
CP0004	C0631	O00406	Delay	App	Medium	2025-13
CP0005	C0535	O00154	Delay	Email	Medium	2024-05
CP0006	C0096	O00147	Delay	App	Medium	2024-07

```
colnames(complaints)
```

```
'complaint_id' · 'customer_id' · 'order_id' · 'complaint_type' · 'channel' · 'severity' · 'created_at' ·  
'status' · 'resolution_days' · 'compensation_amount'
```

```
colSums(is.na(customers))
```

```
customer_id:      0 age:      0 home_zone:      0 customer_type:      0 signup_date:  
      0 loyalty_score:      20 app_engagement_score:      0 preferred_channel:      13  
account_status:      0
```

```
colSums(is.na(deliveries))
```

```
delivery_id:      0 order_id:      0 driver_id:      0 vehicle_id:      0 hub_id:      0  
dispatch_time:      0 delivery_completed_at:      19 delivery_status:      0  
route_distance_km:      0 manual_route_override_count:      0  
proof_of_completion_missing:      0 customer_rating_post_delivery:      14  
fuel_or_charge_cost:      0
```

```
library(dplyr)
```

```
Attaching package: ‘dplyr’
```

```
The following objects are masked from ‘package:stats’:
```

```
filter, lag
```

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
complaints<-distinct(complaints)
```

```
deliveries<-distinct(deliveries)
```

```
str(deliveries)
```

```
tibble [950 × 13] (S3: tbl_df/tbl/data.frame)
 $ delivery_id      : chr [1:950] "DL00001" "DL00002" "DL00003" "
 $ order_id         : chr [1:950] "000938" "000004" "000639" "000
 $ driver_id        : chr [1:950] "D004" "D138" "D006" "D116" ...
 $ vehicle_id       : chr [1:950] "V056" "V007" "V049" "V055" ...
 $ hub_id           : chr [1:950] "H05" "H02" "H02" "H02" ...
 $ dispatch_time    : POSIXct[1:950], format: "2024-06-18 10:57:0
 $ delivery_completed_at : POSIXct[1:950], format: "2024-06-19 09:05:5
 $ delivery_status   : chr [1:950] "Failed" "OnTime" "OnTime" "Del
 $ route_distance_km : num [1:950] 17.26 10.34 7.92 16.42 14.52 ..
 $ manual_route_override_count : num [1:950] 1 1 0 0 1 0 0 1 1 1 ...
 $ proof_of_completion_missing : num [1:950] 0 0 0 0 0 0 0 0 0 ...
 $ customer_rating_post_delivery: num [1:950] 3.07 5 4.98 4.18 4.18 1.57 4.64
 $ fuel_or_charge_cost : num [1:950] 12.05 13.41 8.51 13.62 9.22 ...
```

```
library(readr)
```

```
complaints<-read_csv("complaints.csv")
```

Rows: 320 Columns: 10

— Column specification —

Delimiter: ","

chr (7): complaint_id, customer_id, order_id, complaint_type, channel, severe,

dbl (2): resolution_days, compensation_amount

dtm (1): created_at

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
deliveries<-read_csv("deliveries.csv")
```

Rows: 950 Columns: 13

— Column specification —

Delimiter: ","

chr (6): delivery_id, order_id, driver_id, vehicle_id, hub_id, delivery_status,

dbl (5): route_distance_km, manual_route_override_count, proof_of_completion,

dtm (2): dispatch_time, delivery_completed_at

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
str(deliveries)
```

```
spc_tbl_ [950 × 13] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ delivery_id      : chr [1:950] "DL00001" "DL00002" "DL00003" "
 $ order_id         : chr [1:950] "000938" "000004" "000639" "000
 $ driver_id        : chr [1:950] "D004" "D138" "D006" "D116" ...
 $ vehicle_id       : chr [1:950] "V056" "V007" "V049" "V055" ...
 $ hub_id           : chr [1:950] "H05" "H02" "H02" "H02" ...
 $ dispatch_time    : POSIXct[1:950], format: "2024-06-18 10:57:0
 $ delivery_completed_at : POSIXct[1:950], format: "2024-06-19 09:05:5
 $ delivery_status   : chr [1:950] "Failed" "OnTime" "OnTime" "Del
 $ route_distance_km : num [1:950] 17.26 10.34 7.92 16.42 14.52 ..
 $ manual_route_override_count : num [1:950] 1 1 0 0 1 0 0 1 1 1 ...
 $ proof_of_completion_missing : num [1:950] 0 0 0 0 0 0 0 0 0 0 ...
 $ customer_rating_post_delivery: num [1:950] 3.07 5 4.98 4.18 4.18 1.57 4.64
 $ fuel_or_charge_cost : num [1:950] 12.05 13.41 8.51 13.62 9.22 ...
 - attr(*, "spec")=
 .. cols(
 ..   delivery_id = col_character(),
 ..   order_id = col_character(),
 ..   driver_id = col_character(),
 ..   vehicle_id = col_character(),
 ..   hub_id = col_character(),
 ..   dispatch_time = col_datetime(format = ""),
 ..   delivery_completed_at = col_datetime(format = ""),
 ..   delivery_status = col_character(),
 ..   route_distance_km = col_double(),
 ..   manual_route_override_count = col_double(),
 ..   proof_of_completion_missing = col_double(),
 ..   customer_rating_post_delivery = col_double(),
 ..   fuel_or_charge_cost = col_double()
 .. )
 - attr(*, "problems")=<pointer: 0x5953eb2ae0c0>
```

```
install.packages("sqldf" , repos="http://cloud.r-project.org")
```

Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

also installing the dependencies ‘gsubfn’, ‘proto’, ‘RSQLite’, ‘chron’

```
library(sqldf)
```

Loading required package: gsubfn

Loading required package: proto

Warning message:

“no DISPLAY variable so Tk is not available”

Loading required package: RSQLite

```
complaint_delivery_analysis <- sqldf("
```

```
SELECT
```

```
  c.order_id,
  c.severity,
  d.delivery_status,
  d.route_distance_km,
  d.customer_rating_post_delivery
```

```
FROM complaints c
```

```
JOIN deliveries d
```

```
ON c.order_id = d.order_id
```

```
")
```

```
head(complaint_delivery_analysis)
```

A data.frame: 6 × 5

	order_id	severity	delivery_status	route_distance_km	customer_rating_po
	<chr>	<chr>	<chr>	<dbl>	
1	O00814	High	OnTime	12.88	
2	O00628	Medium	Delayed	12.25	
3	O00384	High	Delayed	9.61	
4	O00406	Medium	OnTime	31.46	
5	O00154	Medium	Delayed	13.63	
6	O00147	Medium	OnTime	4.63	

```
drivers <- read_csv("drivers.csv")
```

Rows: 170 Columns: 8

— Column specification —

Delimiter: ","

chr (4): driver_id, base_zone, employment_type, shift_preference

dbl (4): years_experience, training_score, driver_rating, active_flag

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
head(drivers)
```


A tibble: 6 × 8

driver_id	base_zone	employment_type	years_experience	training_score	driv
<chr>	<chr>	<chr>	<dbl>	<dbl>	
D001	AIRPORT	FullTime	8	67.8	
D002	Central	FullTime	4	42.4	
D003	AIRPORT	FullTime	11	96.5	
D004	Airport	PartTime	13	88.9	
D005	north	FullTime	3	69.7	
D006	CENTRAL	FullTime	8	78.8	

```
driver_delivery_analysis <- sqldf("
```

```
SELECT
```

```
  dr.driver_id,
  d.delivery_status,
  d.route_distance_km,
  d.customer_rating_post_delivery
```

```
FROM drivers dr
```

```
JOIN deliveries d
```

```
ON dr.driver_id = d.driver_id
```

```
")
```

```
head(driver_delivery_analysis)
```

A data.frame: 6 × 4

	driver_id	delivery_status	route_distance_km	customer_rating_post_delive
	<chr>	<chr>	<dbl>	<dbl>
1	D001	Delayed	10.59	3
2	D001	Failed	6.23	4
3	D001	OnTime	9.22	4
4	D001	OnTime	10.11	4
5	D001	OnTime	18.36	3
6	D001	OnTime	21.60	3

```
incidents <- read_csv("incidents.csv")
```

Rows: 280 Columns: 7

— Column specification —

Delimiter: ","

chr (5): incident_id, delivery_id, incident_type, severity, resolution_status

```
dbl (1): resolved_hours
dtm (1): reported_at

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(incidents)
```

A tibble: 6 × 7

incident_id	delivery_id	incident_type	reported_at	severity	resolution_status	resolved_hours
<chr>	<chr>	<chr>	<dtm>	<chr>		
I0001	DL00221	BatteryAlert	2024-03-11 23:46:00	Medium	Escalated	12
I0002	DL00578	BatteryAlert	2024-02-21 10:56:00	Low	Resolved	2
I0003	DL00175	TemperatureIssue	2025-04-17 23:22:00	Medium	Pending	0
I0004	DL00417	ProofMissing	2025-02-09 00:16:00	Medium	Pending	0
I0005	DL00004	CustomerNoShow	2025-01-04 21:26:00	High	Pending	0

```
library(sqldf)
library(readr)
```

```
incidents <- read_csv("incidents.csv")
complaints <- read_csv("complaints.csv")
```

```
Rows: 280 Columns: 7
— Column specification —
Delimiter: ","
chr (5): incident_id, delivery_id, incident_type, severity, resolution_status
dbl (1): resolved_hours
dtm (1): reported_at

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

Rows: 320 Columns: 10
— Column specification —
Delimiter: ","
chr (7): complaint_id, customer_id, order_id, complaint_type, channel, severity, resolution_status
dbl (2): resolution_days, compensation_amount
dtm (1): created_at

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
complaints <- read_csv("complaints.csv")
```

```
Rows: 320 Columns: 10
— Column specification —
```

Delimiter: ","

chr (7): complaint_id, customer_id, order_id, complaint_type, channel, severity,

dbl (2): resolution_days, compensation_amount

dtm (1): created_at

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
incidents <- read_csv("incidents.csv")
```

Rows: 280 **Columns:** 7

— Column specification —

Delimiter: ","

chr (5): incident_id, delivery_id, incident_type, severity, resolution_status

dbl (1): resolved_hours

dtm (1): reported_at

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
incident_complaint_analysis <- sqldf("
```

```
SELECT
```

```
  i.incident_id,
```

```
  i.severity,
```

```
  c.complaint_type,
```

```
  c.status,
```

```
  c.resolution_days
```

```
FROM incidents i
```

```
JOIN deliveries d
```

```
ON i.delivery_id = d.delivery_id
```

```
JOIN complaints c
```

```
ON d.order_id = c.order_id
```

```
LIMIT 10
```

```
")
```

```
head(incident_complaint_analysis)
```

A data.frame: 6 × 5

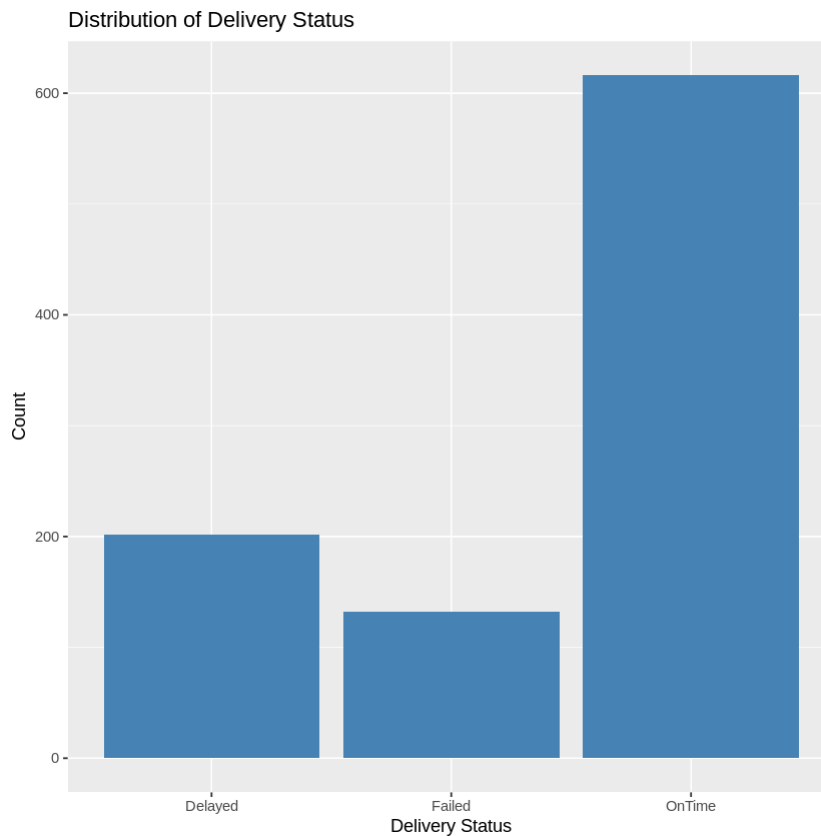
	incident_id	severity	complaint_type	status	resolution_days
	<chr>	<chr>	<chr>	<chr>	<dbl>
1	I0001	Medium	DriverBehaviour	Open	9
2	I0002	Low	SupportExperience	Escalated	3
3	I0009	Medium	AppIssue	Escalated	14
4	I0011	Medium	Delay	Open	15
5	I0011	Medium	MissedPickup	Open	5
6	I0012	Medium	DriverBehaviour	Open	11

```
print(incident_complaint_analysis)
```

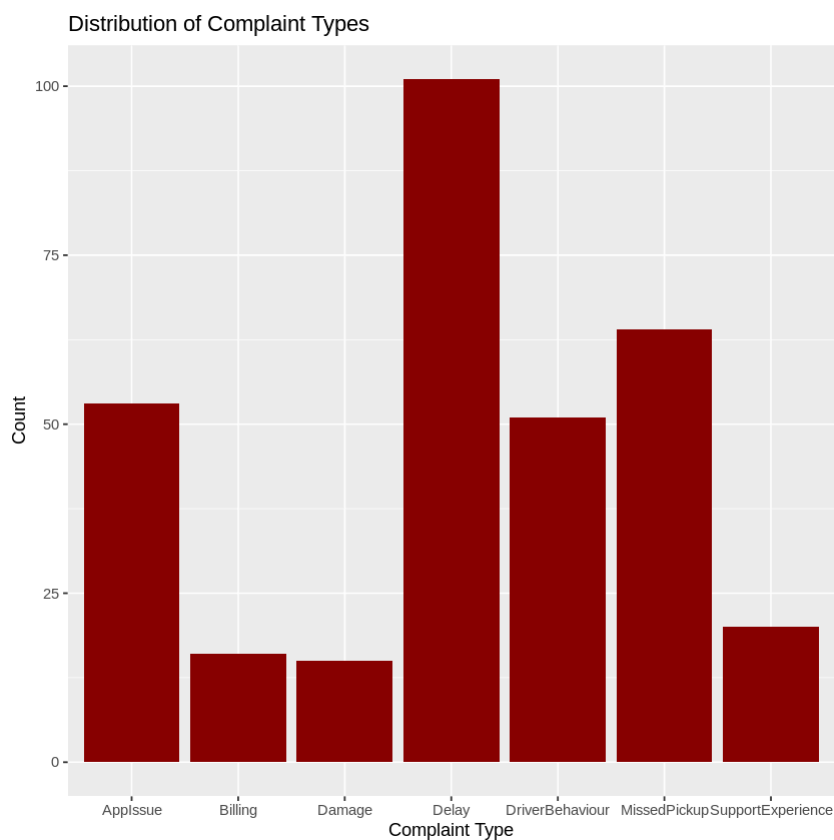
	incident_id	severity	complaint_type	status	resolution_days
1	I0001	Medium	DriverBehaviour	Open	9
2	I0002	Low	SupportExperience	Escalated	3
3	I0009	Medium	AppIssue	Escalated	14
4	I0011	Medium	Delay	Open	15
5	I0011	Medium	MissedPickup	Open	5
6	I0012	Medium	DriverBehaviour	Open	11
7	I0035	Low	MissedPickup	Resolved	1
8	I0035	Low	MissedPickup	Resolved	6
9	I0044	Critical	MissedPickup	Resolved	10
10	I0052	Medium	MissedPickup	Resolved	10

```
library(ggplot2)
```

```
ggplot(deliveries, aes(x = delivery_status)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Distribution of Delivery Status",
    x = "Delivery Status",
    y = "Count"
  )
```



```
ggplot(complaints, aes(x = complaint_type)) +  
  geom_bar(fill = "darkred") +  
  labs(  
    title = "Distribution of Complaint Types",  
    x = "Complaint Type",  
    y = "Count"  
  )
```



```
import pandas as pd
import matplotlib.pyplot as plt
```

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
import pandas as pd

deliveries = pd.read_csv("deliveries.csv")
```

```
deliveries.head()
```

	delivery_id	order_id	driver_id	vehicle_id	hub_id	dispatch_time	deliv
0	DL00001	O00938	D004	V056	H05	2024-06-18 10:57:00	
1	DL00002	O00004	D138	V007	H02	2025-01-11 18:45:00	
2	DL00003	O00639	D006	V049	H02	2025-06-02 20:39:00	
3	DL00004	O00313	D116	V055	H02	2024-03-08 23:31:00	
4	DL00005	O00844	D108	V034	H01	2025-09-21 11:43:00	

```
deliveries["delivery_status"].value_counts()
```

	count
OnTime	616
Delayed	202
Failed	132

dtype: int64

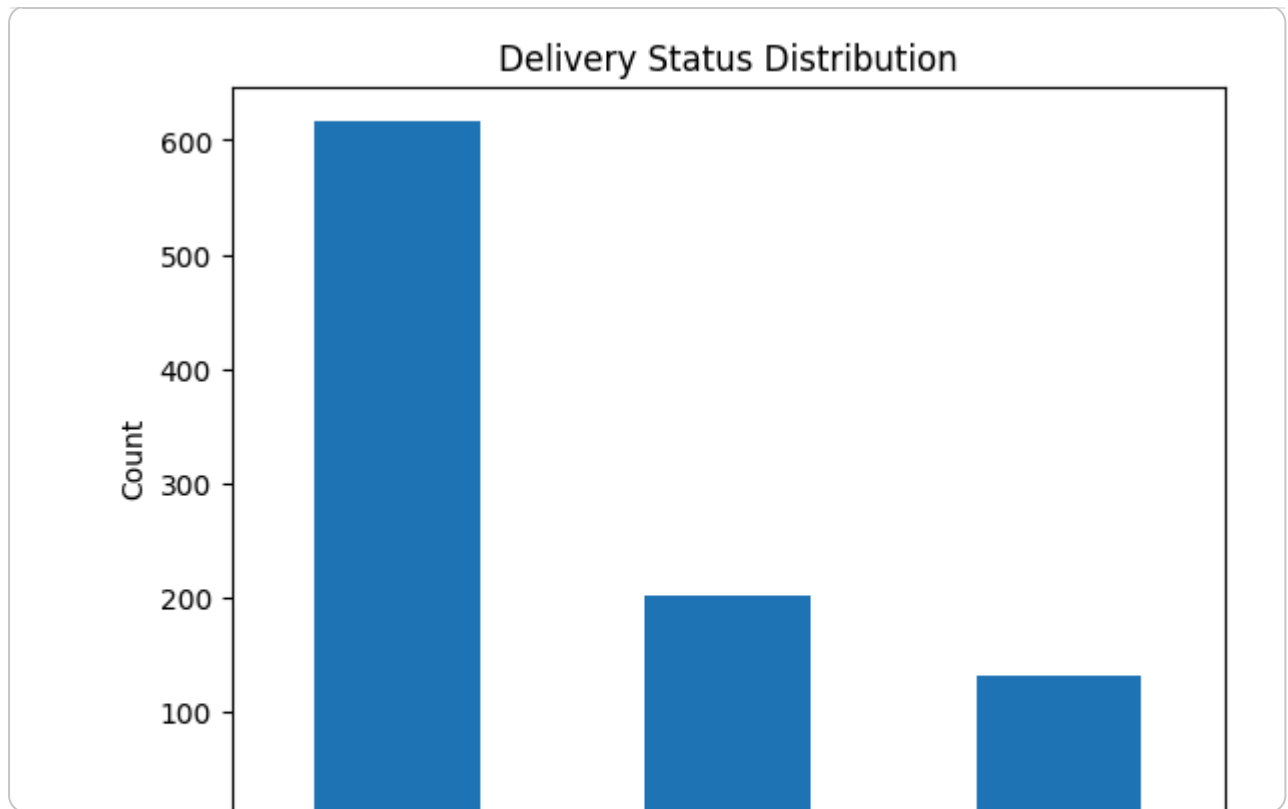
```
deliveries["delivery_status"].value_counts().plot(kind="bar")

plt.title("Delivery Status Distribution")

plt.xlabel("Delivery Status")

plt.ylabel("Count")

plt.show()
```

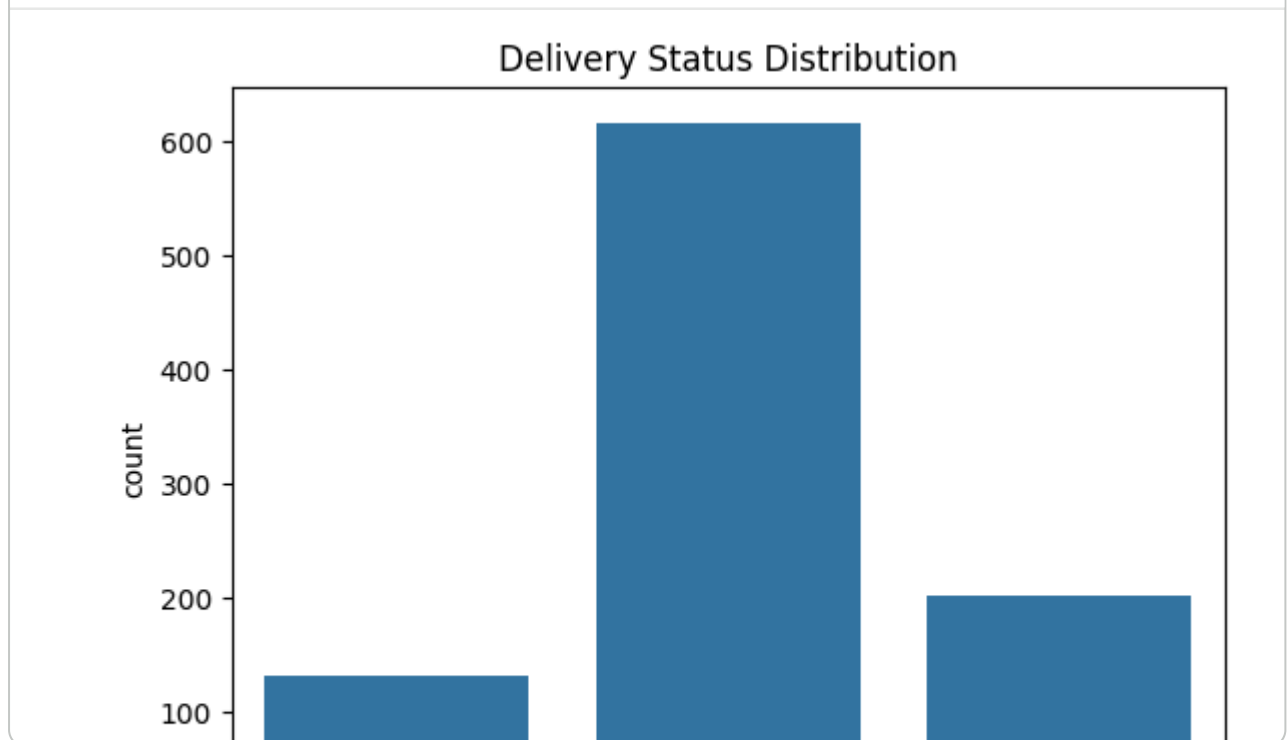


```
import seaborn as sns

sns.countplot(x="delivery_status", data=deliveries)

plt.title("Delivery Status Distribution")

plt.show()
```



```
import pandas as pd

complaints = pd.read_csv("complaints.csv")
```

```
complaints["complaint_type"].value_counts()
```

complaint_type	count
Delay	101
MissedPickup	64
ApplIssue	53
DriverBehaviour	51
SupportExperience	20
Billing	16
Damage	15

dtype: int64

```
complaints["complaint_type"].value_counts().plot(kind="bar")

plt.title("Complaint Type Distribution")

plt.xlabel("Complaint Type")

plt.ylabel("Count")

plt.show()
```

